

Hunters Hill High School
Mathematics
Trial Examination, 2017



Hunters Hill

High School

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- The marks for each question are shown on the paper
- Show all necessary working in questions 11-16

Total Marks: 100

Section I Pages 3-5
10 marks

- Attempt Questions 1-10
- Allow about 20 minutes for this section

Section II Pages 6-12
90 marks

- Attempt Questions 11-16
- Allow about 2 hours and 40 minutes for this section

Section I**10 marks Attempt Questions 1–10****Allow about 20 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1. What is 2.807956 correct to 3 significant figures?

- (A) 2.80
- (B) 2.81
- (C) 2.807
- (D) 2.808

2. A card is drawn at random from a standard deck of playing cards .
What is the probability that the card drawn is a Diamond OR a Queen?

- (A) $\frac{13}{52}$
- (B) $\frac{4}{52}$
- (C) $\frac{17}{52}$
- (D) $\frac{16}{52}$

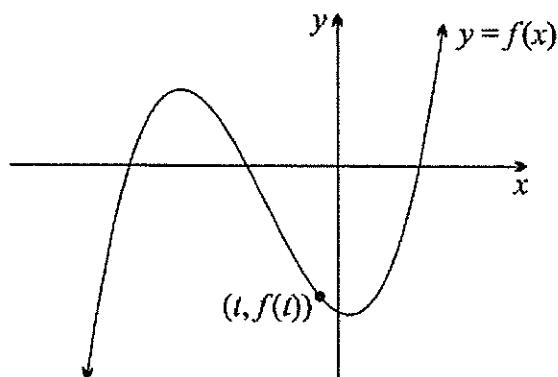
3. The equation $y = 3 \cos(\pi x - 3)$ has a period of:

- (A) 2π
- (B) π
- (C) 2
- (D) 3

4. When $\frac{3 + \sqrt{2}}{3 + 2\sqrt{2}}$ is expressed in the form $a - \sqrt{b}$, then:

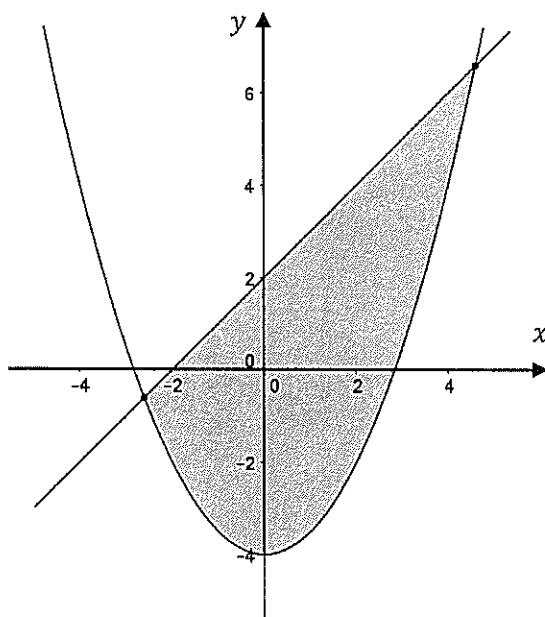
- (A) $a = 1, b = \frac{1}{4}$
- (B) $a = 9, b = 50$
- (C) $a = 5, b = 2$
- (D) $a = 5, b = 18$

5. The diagram shows the graph of $y = f(x)$. Which of the following statements is true?



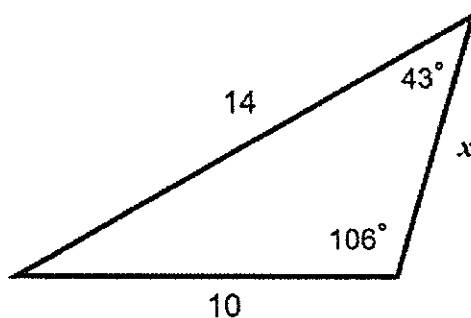
- (A) $f'(t) < 0$ and $f''(t) > 0$
(B) $f'(t) < 0$ and $f''(t) < 0$
(C) $f'(t) > 0$ and $f''(t) < 0$
(D) $f'(t) > 0$ and $f''(t) < 0$
6. The maximum value of the expression $-2x^2 - 4x + 7$ is:
- (A) -1
(B) 1
(C) 7
(D) 9
7. For what domain and range is the function $y = \frac{1}{\sqrt{x-4}}$ defined?
- (A) Domain: $x \geq 4$, Range: $y > 0$
(B) Domain: $x > 4$, Range: $y > 0$
(C) Domain: all real x , Range: all real y
(D) Domain: $x < -2, x > 2$, Range: $y < 0$
8. The values of k for which $x^2 - kx + k + 3 = 0$ has no real roots are:
- (A) $k < 0$
(B) $-2 < k < 6$
(C) $k < -2, k > 6$
(D) $k = 2, 6$

9. Which statement is consistent with the region shown?



- (A) $2y \geq x^2 - 8$ and $x - y + 2 \geq 0$
- (B) $2y \leq x^2 - 8$ and $x - y + 2 \geq 0$
- (C) $2y \geq x^2 - 8$ and $x - y + 2 \leq 0$
- (D) $2y \leq x^2 - 8$ and $x - y + 2 \leq 0$

10. For the triangle below, which of the following statements is true?



- (A) $x = \sin 43^\circ \cdot \frac{14}{\sin 105^\circ}$
- (B) $x = \sin 31^\circ \cdot \frac{14}{\sin 43^\circ}$
- (C) $x = \sin 31^\circ \cdot \frac{10}{\sin 43^\circ}$
- (D) $x = \sin 74^\circ \cdot \frac{10}{\sin 43^\circ}$

End of Section I

Section II**90 marks****Attempt Questions 11–16****Allow about 2 hours and 40 minutes for this section****Begin each question on a NEW SHEET of paper.**

In questions 11 – 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 – Use a SEPARATE writing booklet.**(15 marks)**

- a. Solve $|x - 7| \geq 3$ 2
- b. Factorise $x^3 - 5x$ completely 2
- c. Simplify $\sqrt{5} + \sqrt{20} - 2\sqrt{45}$ 2
- d. Integrate
- i. $2 \sin 6x$ 2
- ii. $2e^x + \frac{1}{x}$ 2
- e. Solve simultaneously 2
- $3x - 2y = 5$
 $x + 3y = 9$
- f. Find the equation of the tangent to the curve $y = x^2 - 5x + 1$ at the point $(1, -3)$. 2
- g. Express $0.\dot{1}\dot{3}$ as a fraction, showing necessary working. 1

End of Question 11

Question 12 – Use a SEPARATE writing booklet.

(15 marks)

a. Evaluate**2**

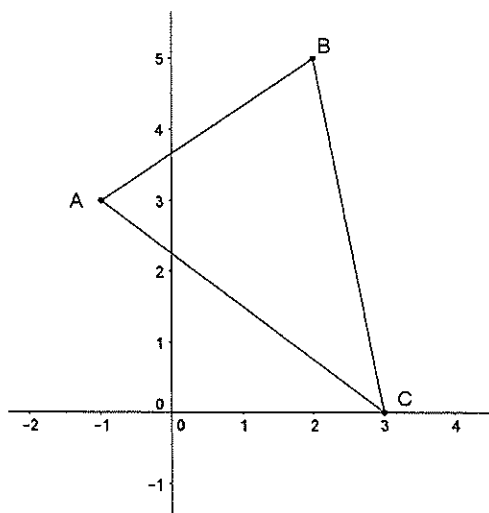
$$\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$$

b. Differentiate

i. $e^x \cos x$

2

ii. $(1 - \sin 2x)^4$

2**c. Find the discriminant of the quadratic equation, $3x^2 + 6x - 1 = 0$, and hence describe the roots of the equation.****2****d. In the diagram below, the three points $A(-1, 3)$, $B(2, 5)$ and $C(3, 0)$ form the vertices of a triangle.****i. Find the length of the interval BC .****1****ii. Show that the equation of BC is $5x + y - 15 = 0$.****1****iii. Find the perpendicular distance from A to the line BC .****1****iv. Hence, find the area of the triangle ABC .****1****Question 12 continues on next page**

- e. An archer fires three arrows at a target. The probability of any single arrow hitting the target is $\frac{4}{5}$.

Find the probability that:

- | | |
|--|---|
| i. The first arrow hits and the next two miss. | 1 |
| ii. The archer hits the target exactly once. | 1 |
| iii. The archer hits the target at least once. | 1 |

End of Question 12

Question 13 – Use a SEPARATE writing booklet.

(15 marks)

a. Find the equation of the normal to the curve $y = 2 \cos x$ when $x = \frac{\pi}{4}$. 3

b. Explain why 1

$$\int_{-3}^3 (x^2 + 4) dx = 2 \int_0^3 (x^2 + 4) dx$$

c. Given that $\log_5 6 = 1.11$ and $\log_5 3 = 0.68$, find:

- | | | |
|------|---------------------|---|
| i. | log ₅ 18 | 1 |
| ii. | log ₅ 2 | 1 |
| iii. | log ₅ 20 | 1 |

d. Solve 3

$$(x + 3)^2 + 5(x + 3) - 14 = 0$$

e. Given that the quadratic equation $2x^2 - 5x + 6 = 0$ has roots of α and β , find the values of:

- | | | |
|-----|--------------------------------------|---|
| i. | α + β | 1 |
| ii. | $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |

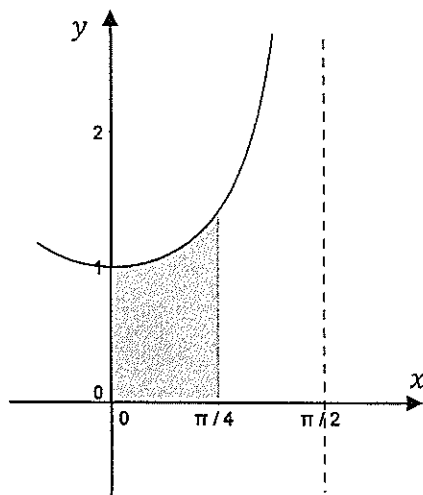
f. Given that $\tan \theta = -\frac{1}{4}$ and θ is a reflex angle, find the value of $\sin \theta$. 2

End of Question 13

Question 14 – Use a SEPARATE writing booklet.

(15 marks)

- a. Find the volume of the solid formed when the area between the curve $y = \sec x$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{4}$, is rotated about the x -axis. 3



- b. i. Express $y = \log_3 x$ with a base of e . 1
- ii. Hence, find $\frac{d}{dx}(\log_3 x)$. 1
- c. Given the cubic function, $y = x^3 - 3x^2 + 2x$
- i. Factorise completely to find the x -intercepts. 2
- ii. Find the location and nature of the stationary points. 3
- iii. Show that the function has a point of inflexion at $(1, 0)$. 1
- iv. Sketch the cubic function, showing all relevant information. 2
- d. A 2L bucket is being filled with water at a rate, $R = 20 + 2t \text{ cm}^3 \text{ s}^{-1}$.
- i. If it initially contains 500 cm^3 of water, find an expression for the volume of water in the bucket. 1
- ii. Show that the bucket is full after 30 seconds. 1

End of Question 14

Question 15 – Use a SEPARATE writing booklet.

(15 marks)

- a. On a small island, an ecologist monitors a colony of rabbits. When first assessed, there are 120 rabbits.

The number of rabbits, $N(t)$, after t years is given by

$$N(t) = 120e^{kt}.$$

- i. After 4 years there are 280 rabbits.
Show that $k = 0.2118$, correct to four decimal places. 1
- ii. How many rabbits are in the colony when $t = 6$? 1
- iii. What is the rate of change of the number of rabbits per year when $t = 6$? 1
- iv. How long does it take for the number of rabbits to increase from 120 to 500? 2

- b. Evaluate 3

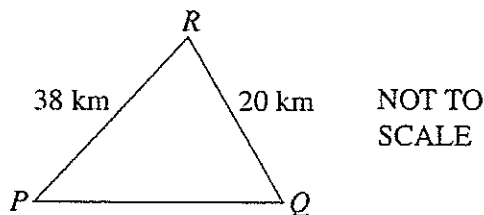
$$\int_0^2 \frac{x}{3x^2 + 4} dx$$

- c. Consider the function $y = \ln(x + 1)$ for $x > -1$.

- i. Sketch the function, showing its essential features. 1
- ii. Use Simpson's rule with three function values to find an approximation to 2

$$\int_0^4 \ln(x + 1) dx.$$

- d. In the diagram below, the point Q is due east of P . The point R is 38 km from P and 20 km from Q . The bearing of R from Q is 315° .



- i. What is the size of $\angle PQR$? 1
- ii. What is the bearing of R from P ? 3

End of Question 15

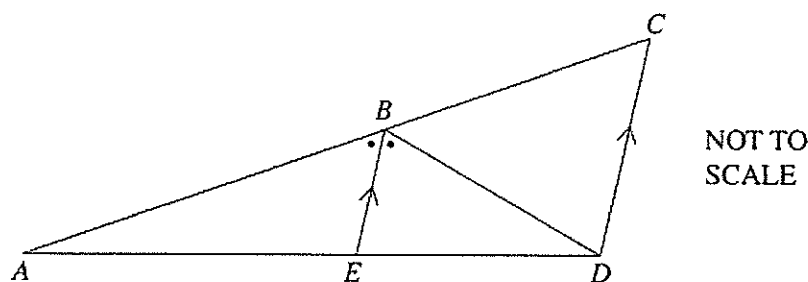
Question 16 – Use a SEPARATE writing booklet.

(15 marks)

- a. A particle moves along a straight line so that its displacement, x metres, from a fixed point O is given by $x = 1 + 3 \cos 2t$, where t is measured in seconds.

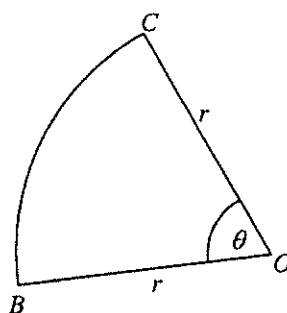
- i. What is the initial displacement of the particle? 1
- ii. Sketch the graph of x as a function of t for $0 \leq t \leq \pi$. 2
- iii. Hence, or otherwise, find when the particle first comes to rest after $t = 0$. 1
- iv. Find a time when the particle reaches its maximum speed.
What is this speed? 2

- b. In the diagram below, $BE \parallel CD$ and BE bisects $\angle ABD$.



- i. Explain why $\angle EBD = \angle BDC$. 1
- ii. Prove that $\triangle BCD$ is isosceles. 2

- c. The diagram below shows a sector OBC of a circle with centre O and radius r cm. The arc BC subtends an angle θ radians at O .



- i. Show that the perimeter of the sector is $P = r(2 + \theta)$ 1
- ii. Given that the perimeter of the sector is 36 cm, show that its area is given by 2

$$A = \frac{648\theta}{(\theta + 2)^2}$$
- iii. Hence, show that the maximum area of the sector is 81 cm^2 . 3

End of paper

2017 TRIAL - H4HS - MAT.

SOLUTIONS

SECTION I

1. B 6. D
2. D 7. B
3. C 8. B
4. D 9. A
5. A 10. C

SECTION II

a) $|x-7| \geq 3$.

$x-7 \leq -3, \quad x-7 \geq 3$

$x \leq 4, \quad x \geq 10$

1 - Resolving abs value
1 - directions

b) $x^8 - 5x = x(x^7 - 5)$

$= x(x+\sqrt{5})(x-\sqrt{5})$

1 - common factor
1 - diff 2sg

c) $\sqrt{5+120-2\sqrt{45}} = \sqrt{5+2\sqrt{5}-2 \cdot 3\sqrt{5}}$

$= -3\sqrt{5}$

1 - simplify to $\sqrt{5}$
1 - sum

d) i. $\int 2 \sin 6x = 2 \left(\frac{-\cos 6x}{6} \right) + c$

$= -\frac{1}{3} \cos 6x + c$

1 - integral
1 - +c

ii. $\int \left(2e^x + \frac{1}{x} \right) dx = 2e^x + \ln x + c$

1 - $\int e^x$
1 - $\int \frac{1}{x}$

e) $3x - 2y = 5$

$x + 3y = 9$

ii x3 $3x + 9y = 27$

1 - popes
1 - answers

iii-i $11y = 22$

$y = 2$

Sub into ii

$x + 3(2) = 9$

$x = 3$

test in i

$LHS = 3(3) - 3(2)$

$= 9 - 6$

$= 3 = RHS$

$\therefore x=3, y=2$ is the solution

f) $y = x^2 - 5x + 1$

$y' = 2x - 5$

at $(1, -3)$

$m = 2(1) - 5$

$= -3$

1 - equation

by point-gradient,

$y - (-3) = -3(x - 1)$

$\therefore 3x + y = 0$ is the equation of the tangent.

g) let $n = 0.13$

$100n = 13 \cdot 13$

ii-i

$99n = 13$

$\therefore n = \frac{13}{99}$

99

1 - correct process

Question 12

a) $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x} = \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin x}{\sin 2x}$ | - numerator of $\sin x$

$= \frac{3}{2} \times 1$

$= \frac{3}{2}$

b) $(1 - \sin 2x)$

i) $\frac{d}{dx}(e^x \cos x) = e^x \cos x + e^x (-\sin x)$ | - correct $\frac{d}{dx}$
 $= e^x (\cos x - \sin x)$ | - product rule

ii) $\frac{d}{dx} (1 - \sin 2x)^4 = 4(1 - \sin 2x)^3 (-2 \cos 2x)$ | - chain rule
 | - ans

c) $3x^2 + 6x - 1 = 0$

$\Delta = 6^2 - 4(3)(-1)$ | - Δ
 $= 36 + 12$ | - description
 $= 48$

$\therefore$ roots are real and distinct.

d) ii) $P(2.5), C(3.0)$ | - ans
 by two-point formula.

$y - 0 = \frac{5 - 0}{2 - 3} (x - 3)$

$y = -5(x - 3)$
 $\therefore 5xy - 15 = 0$ is the equation of BC.

i) $d_{BC} = \sqrt{(3-2)^2 + (0-5)^2}$
 $= \sqrt{1 + 25}$
 $= \sqrt{26}$ | - ans

ii) $d \cdot A(-1, 3)$ BC. $5x + y - 15 = 0$

$d = \frac{|5(-1) + (3) - 15|}{\sqrt{5^2 + 1^2}}$ | - ans

$= \frac{|-17|}{\sqrt{26}} = \frac{17}{\sqrt{26}}$

ii) $A = \frac{1}{2} b h$ | - ans
 $= \frac{1}{2} \sqrt{26} \cdot \frac{17}{\sqrt{26}}$
 $= \frac{17}{2}$ units²

$$e) i) P(L.H) = \frac{4}{5}, P(R.H) = \frac{1}{5}$$

$$P(H.M.N) = \frac{4}{5} \cdot \frac{1}{5} \cdot \frac{1}{5}$$

$$= \frac{4}{125}$$

$$ii) P(\text{one H}) = \frac{4}{5} \times 3$$

$$= \frac{12}{5}$$

$$iii) P(\text{at least one H}) = 1 - P(\text{B.M})$$

$$= 1 - \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5}$$

$$= \frac{124}{125}$$

Question 13

$$a) y = 2 \cos x$$

$$y = -2 \sin x$$

$$\text{at } x = \frac{\pi}{4}$$

$$y = 2 \cos \frac{\pi}{4}$$

$$= 2 \cdot \frac{1}{\sqrt{2}}$$

$$= \sqrt{2}$$

$$y = -2 \sin \frac{\pi}{4}$$

$$= -2 \cdot \frac{1}{\sqrt{2}}$$

$$= -\sqrt{2}$$

$$M_u = -\frac{1}{M_T}$$

$$= \frac{\sqrt{2}}{2}$$

by point gradient

$$y - \sqrt{2} = \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4} \right)$$

$$8y - 2\sqrt{2} = 4\sqrt{2}x - \pi\sqrt{2}$$

$$\therefore 4\sqrt{2}x - 8y + 12(8 - \pi) = 0 \text{ is the equation of normal}$$

b.) $f(x) = x^2 + 4$ is an even function and the integral has opposite or equal bounds

$$c) i) \log_5 6 = 1.11 \quad \log_5 3 = 0.68$$

$$\log_5 18 = \log_5 3 + \log_5 6$$

$$= 0.68 + 1.11$$

$$= 1.79$$

$$ii) \log_5 2 = \log_5 6 - \log_5 3$$

$$= 1.11 - 0.68$$

$$= 0.43$$

$$\text{iii) } \log_5 20 = \log_5 5 + \log_5 4$$

$$= \log_5 5 + 2 \log_5 2$$

$$= 1 + 2 \times 0.43$$

$$= 1.86$$

$$\text{d) } (x+3)^2 + 5(x+3) - 14 = 0$$

$$\text{let } u = x+3$$

$$u^2 + 5u - 14 = 0$$

$$(u+7)(u-2) = 0$$

$$\therefore u = -7, 2$$

$$x+3 = -7, 2$$

$$\therefore x = -10, -1$$

$$\text{e) } 2x^2 - 5x + 6 = 0$$

$$\text{i) } x + \beta = -\frac{(-5)}{2}$$

$$= \frac{5}{2}$$

$$\text{ii) } \alpha\beta = \frac{6}{2}$$

$$= 3$$

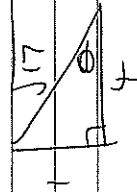
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{5}{3}$$

$$= \frac{5}{6}$$

$$\text{f) } \tan \theta = -\frac{1}{4}$$

⊙ is reflex, so lies in 3rd or 4th quadrant
 $\tan \theta < 0$ so θ in 3rd quadrant



$$\therefore \sin \theta = -\frac{1}{\sqrt{17}}$$

Question 14.

a)

$$y = \sec x$$

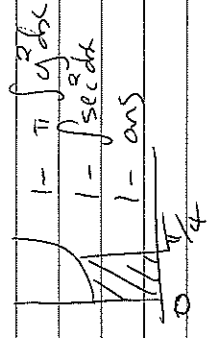
$$V = \pi \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sec^2 x \, dx$$

$$= \pi \left[\tan x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \pi \left(\tan \frac{\pi}{2} - \tan \frac{\pi}{4} \right)$$

$$= \pi (1 - 0)$$

$$= \pi \text{ units}^3$$



$$\text{b) i) } y = \log_3 x$$

$$y = \frac{\ln x}{\ln 3}$$

$$\text{ii) } \frac{d}{dx} (\log_3 x) = \frac{d}{dx} \left(\frac{\ln x}{\ln 3} \right)$$

$$= \frac{1}{\ln 3} \frac{d}{dx} (\ln x)$$

$$= \frac{1}{\ln 3} \cdot \frac{1}{x}$$

$$= \frac{1}{x \ln 3}$$

ii) $y'' = 6x - 6$

1 - shown

for inflection, $y'' = 0$

$6x - 6 = 0$

$x = 1$

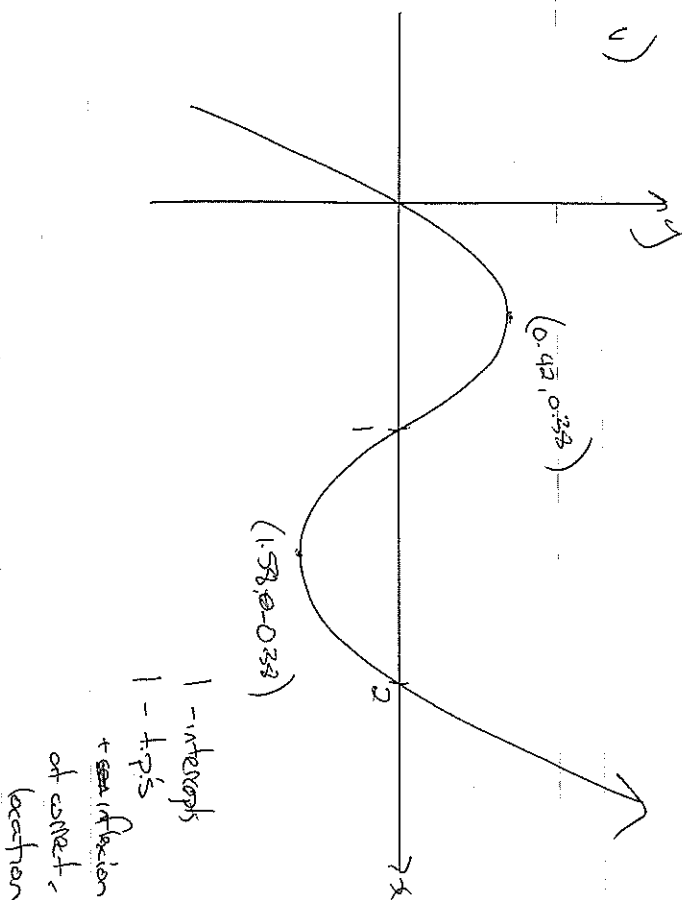
$y = 1^3 - 3(1)^2 = 2(1)$

test for inflection

$= 0$

x	0	1	2
y''	-6	0	6
	< 0	0	> 0

as concavity changes, inflection at $(1, 0)$



c) $y = x^3 - 3x^2 + 2x$

i) $y = x(x^2 - 3x + 2)$

$= x(x-2)(x-1)$

for x-int, let $y = 0$

$x(x-1)(x-2) = 0$

$x = 0, 1, 2$ are intercepts

ii) $y' = 3x^2 - 6x + 2$

for stationary points, $y' = 0$

$3x^2 - 6x + 2 = 0$

$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(2)}}{2(3)}$

$= \frac{6 \pm \sqrt{12}}{6}$

$= \frac{3 \pm \sqrt{3}}{3} \left(\approx 1.58, 0.42 \right)$

$y'' = 6x - 6 = 1 \pm \frac{\sqrt{3}}{3}$

for $x = \frac{3 - \sqrt{3}}{3} (0.42)$

for $x = \frac{3 + \sqrt{3}}{3} (1.58)$

$y \approx 0.42^3 - 3(0.42)^2 + 2(0.42)$
 $= 0.38$

$y \approx 1.58^3 - 3(1.58)^2 + 2(1.58)$
 $= -0.38$

$y'' \approx 6(0.42) - 6 < 0$

$y'' \approx 6(1.58) - 6 > 0$

$\therefore$ stationary points at $(0.42, 0.38)$ and $(1.58, -0.38)$ are a local maximum and local minimum, respectively

d) $R = 20 + 2t \text{ cm}^3 \text{ s}^{-1}$

i) $t=0, V=500$

$$V = \int R dt$$

$$= \int (20 + 2t) dt$$

$$= 20t + t^2 + c$$

when $t=0, V=500$

$$500 = 20(0) + 0 + c$$

$$c = 500$$

so $V = 20t + t^2 + 500 \text{ cm}^3$

ii) at $t=30$

$$V = 20(30) + 30^2 + 500$$

$$= 600 + 900 + 500$$

$$= 2000 \text{ cm}^3$$

$$= 2 \text{ L}$$

$\therefore$ bucket is full after 30 sec

Question 15

a) $N(t) = 120e^{kt}$, $t=0, N(t)=120$

i) at $t=4, N(t)=280$

$$\therefore 280 = 120e^{4k}$$

$$e^{4k} = \frac{280}{120}$$

$$= \frac{7}{3}$$

$$4k = \ln\left(\frac{7}{3}\right)$$

$$k = \frac{1}{4} \ln\left(\frac{7}{3}\right)$$

$$= 0.2118244651$$

$$= 0.2118 \text{ (4dp)}$$

ii) $N(t) = 120e^{0.2118t}$

1-ans

$$N(6) = 120e^{0.2118 \times 6}$$

$$= 427.642861$$

$$\text{(or } 427.7070649 \text{)}$$

$\therefore$ there are 427 rabbits when $t=6$

iii) $\frac{dN(t)}{dt} = 120e^{0.2118t} (0.2118)$

$$\frac{dN(6)}{dt} = 120e^{0.2118(6)} (0.2118)$$

1-ans

$$= 90.59882023 \text{ (or } 90.5750598 \text{)}$$

at $t=6$, the rate of change ~~of~~ is 90.60 rabbits/year. (2dp)

$$10) \quad N(t) = 120e^{-0.2118t}$$

$$\text{for } N(t) = 500$$

$$120e^{-0.2118t} = 500$$

$$e^{-0.2118t} = \ln(500/120)$$

$$+ = \frac{\ln(500/120)}{-0.2118}$$

$$= 6.737259333 \quad (\sim 6.74 \text{ yrs})$$

it will take 6.74 years to reach 500 rabbits

[in the 7th year]

$$b) \quad \int_0^2 \frac{7x}{3x^2+4} dx = \frac{1}{6} \int_0^2 \frac{6x}{3x^2+4} dx$$

$$= \frac{1}{6} \left[\ln(3x^2+4) \right]_0^2$$

$$= \frac{1}{6} (\ln(3(2)^2+4) - \ln(3(0)^2+4))$$

$$= \frac{1}{6} (\ln 16 - \ln 4)$$

$$= \frac{1}{6} \ln 4$$

$$= 0.2310490602$$

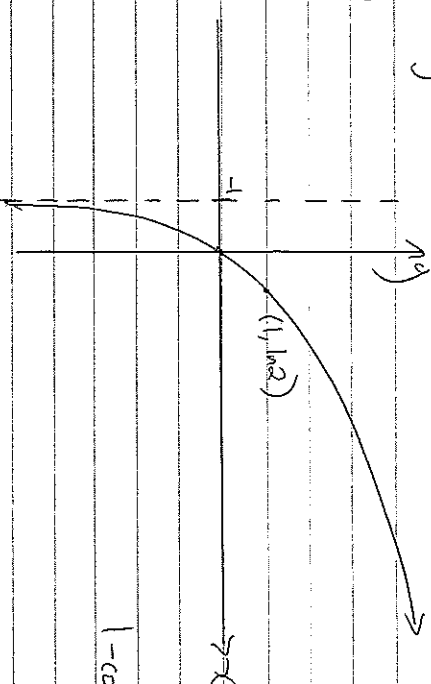
1 - value.

1 - $\int \rightarrow \log$.

1 - $x \frac{6}{6}$

$$c) \quad y = \ln(x+1), \quad x > -1$$

i)



1 - correct + labelled

$$ii) \quad \begin{array}{c|ccc} x & 0 & 2 & 4 \\ \hline \ln(x+1) & \ln 1 & \ln 3 & \ln 5 \end{array}$$

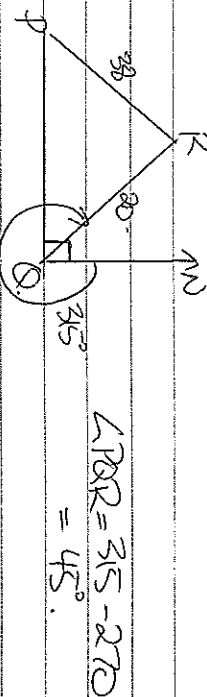
$$\int_0^4 \ln(x+1) dx \doteq \frac{4-0}{6} \left[\ln 1 + 4 \ln 3 + \ln 5 \right]$$

$$= \frac{2}{3} [\ln 3^4 + \ln 5]$$

$$= 2 \ln 324$$

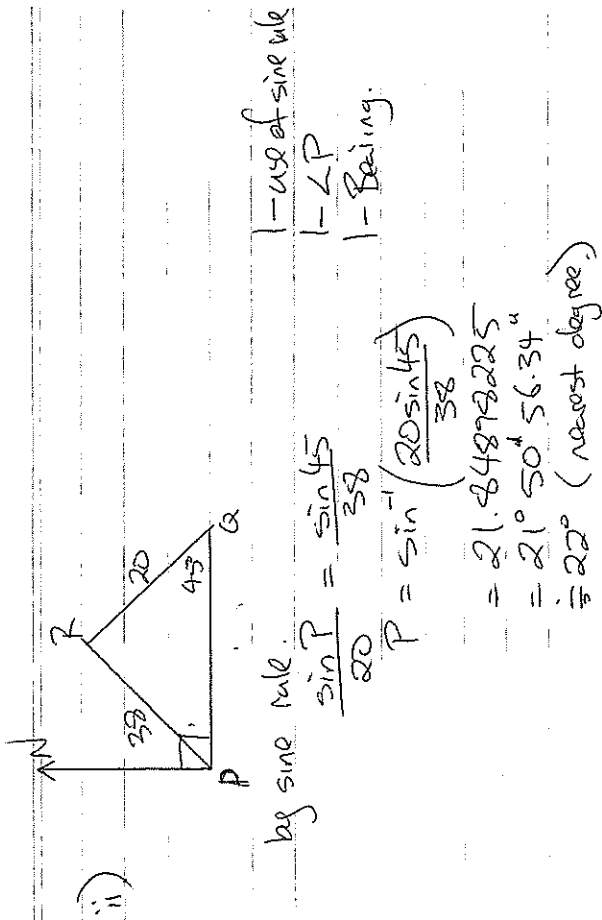
$$= 4.002591378 \quad \text{1 - use of Simpson}$$

d) ii)



$$\angle PRQ = 315 - 270 = 45^\circ$$

1 - ans



$\angle NPR = 90 - 22 = 68^\circ$

Bearing of R from P is $068^\circ T$

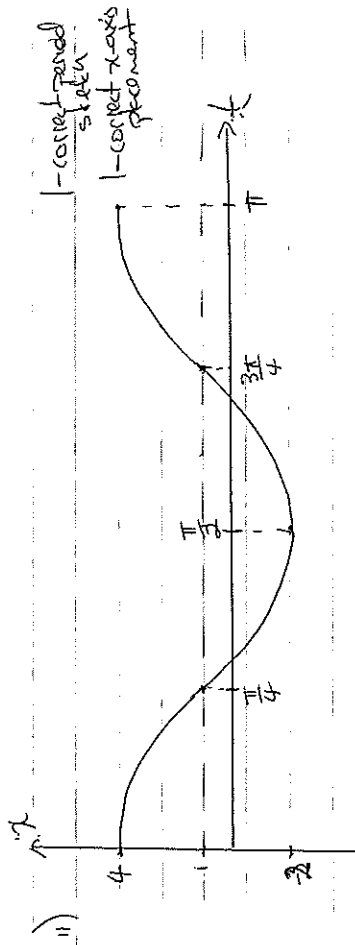
Question 16

a) $x = 1 + 3 \cos 2t$

i) when $t = 0$

$x = 1 + 3 \cos 2(0)$
 $= 1 + 3(1)$
 $= 4$

$\therefore$ initial displacement is 4 units



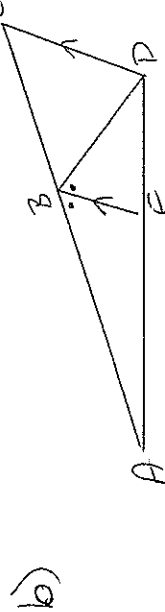
iii) Particle comes to rest at $\frac{\pi}{2}$ sec. 1-ans

iv) Max speed at $t = \frac{\pi}{4}$

$\dot{x} = -6 \sin 2t$
 $= -6 \sin 2\left(\frac{\pi}{4}\right)$
 $= -6 \sin \frac{\pi}{2}$
 $= -6$

1-time
1-speed

max speed is 6 ms^{-1}



i) alternate angles in parallel lines, $EB \parallel DC$, are equal
 - reason

ii) $\angle BCD = \angle ABE$ (corresponding angles in parallel lines, $EB \parallel DC$)
 $= \angle EBD$ (BE bisects $\angle ABD$)
 $= \angle BDC$ (opposite angles)
 $\therefore \angle BCD = \angle BDC$
 1- $\angle BCD = \angle ABE$ + reason
 1-complete proof

$\therefore \triangle BCD$ is isosceles (pair of equal angles)

i) $P = r + r + r\theta$
 $= 2r + r\theta$
 $= r(2 + \theta)$
 - shown

ii) $P = 36$
 $\therefore r(2 + \theta) = 36 \Rightarrow r = \frac{36}{2 + \theta}$

$A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times \left(\frac{36}{2 + \theta} \right)^2 \theta$ | - expression for r
 $= \frac{1296 \theta}{2(2 + \theta)^2}$ | - subst. into A and show
 $= \frac{648 \theta}{(2 + \theta)^2}$

iii) $\frac{dA}{d\theta} = \frac{(2 + \theta)^2 \cdot 648 - 648 \theta \cdot 2(2 + \theta)}{(2 + \theta)^4}$ | - quotient rule
 $= \frac{648(2 + \theta)^2 [(2 + \theta) - 2\theta]}{(2 + \theta)^4}$
 $= \frac{648(2 + \theta)(2 - \theta)}{(2 + \theta)^3}$ | - use of quotient rule
 $= \frac{648(2 - \theta)}{(2 + \theta)^3}$ | - value of θ

for max area, $\frac{dA}{d\theta} = 0$ | - Next shown

$\therefore 648(2 - \theta) = 0$
 $\theta = 2$

$A = \frac{1}{2} \cdot \frac{648(2)}{(2 + 2)^2}$
 $= 81$

Hence, maximum area of sector is 81 cm^2

test gradient for next point,

$\frac{dA}{d\theta}$	0	0	2	3
	0	162	0	-5.184